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3D thermal model to investigate component displacement phenomenon during reflow soldering

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ABSTRACT

This paper reports the development of a 3D component-level thermal model to investigate the displacement of the components during the reflow soldering process. One of the root causes of displacement is the temperature deviation between the contact surfaces of the components. Therefore, our model examines the temperature distribution at the level of discrete components and not at the level of whole assembly as is the case of general models used in reflow soldering [Whalley DC. A simplified reflow soldering process model. J Mater Process Technol 2004;150:134–44; Sarvar F, Conway PP. Effective modelling of the reflow soldering process: basis construction and operation of a process model. IEEE Trans Compon Pack Manuf Technol C: Manuf 1998;21(2):126–33; Inoue M, Koyanagawa T. Thermal simulation for predicting substrate temperature during reflow soldering process. In: IEEE Proceeding of the 55th ECTC; 2005. p. 1021–6].

Our model is based on the thermal or central node theory. This means that the investigated area is divided into thermal cells representing the thermal behaviour of the given material. During the model discretization a nonuniform cuboid cell grid is used, with this the resolution of the model can be refined to focus on areas of interest (e.g. the soldering pads) whilst reducing the involvement with the less important areas (such as the epoxy body of a component). This approach therefore allows to achieve a significant increase of spatial resolution of the calculated temperature distribution in some distinct areas with only a little increase of model complexity. The model is described by common heat conduction and convection equations and these are solved by the finite difference method in order to achieve high calculation speeds and easy implementation. The ability of our single purpose model is then compared with a general purpose FEM analyzer. It results that the calculation accuracy of our model is comparable with more detailed models. Also, its calculation speed and application is much faster and easier.

1. Introduction

Reflow soldering is an important step of surface mounted technology (SMT). This process is applied to enable the attachment of surface mount devices (SMDs) to printed wiring boards (PWBs). The preparatory steps of the process are the solder paste printing to the contact surfaces (pads) of the PWB, and the component placement onto the solder paste deposit. The reflow process then heats the entire assembly to a temperature beyond the melting point (reflow temperature) of the solder alloy. This allows the melting of the individual solder particles in the paste into a single volume which can wet the soldering surfaces and form the solder joints. Usually the heating of the assembly is achieved by a forced convection oven containing a number of independently controllable heating zones [1]. The assembly is transported through the oven by a conveyor.

It was reported in [4–6] that the heat flux transferred by forced convection differs significantly at different component faces. In addition the conduction properties of the components are not always homogeneous. Therefore, it is possible that the temperature gradient can also be large in a component and not only in the whole assembly. This can generate reflow soldering failures such as component displacement but this usually occurs in the case of big components. The formation of the component displacement is the following: the inhomogeneous temperature distribution in the component can cause a delay in the start of the melting of the solder between the individual solder pads. This effect generates wetting force differences between the solder pads which can easily move the whole component during the soldering and causes the displacement phenomenon. The component displacement is a serious problem. It can result in electrical shorts, open joints and also decrease the reliability of the joints themselves.

For our case study, we have chosen to investigate the displacement of the TO263 package (Fig. 1). These packages have between 2 and 7 terminals and mainly contain a power application which is placed on a heat sink (Fig. 2). Terminals are connected to the chip

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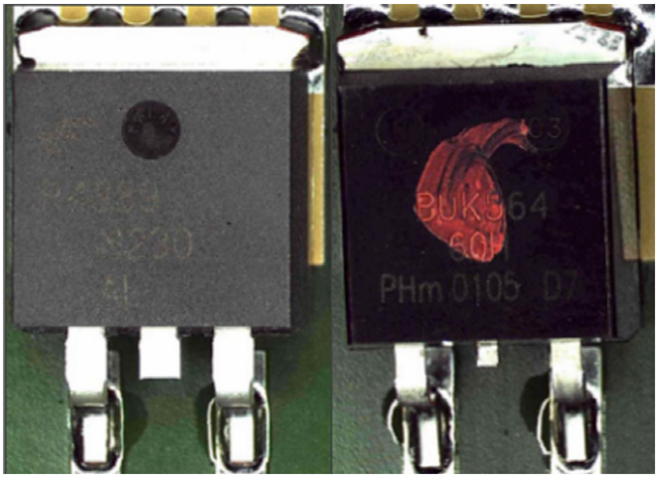

Fig. 1. Left: TO263 package (dimensions: $10 \times 9 \times 6$ mm^3); Right: Displaced TO263.

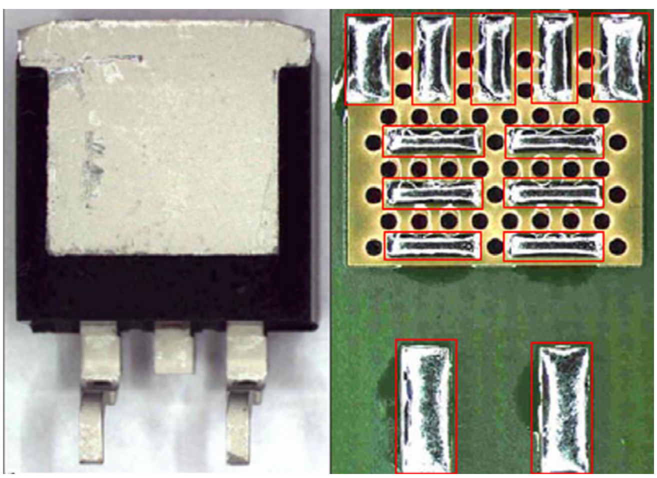

Fig. 2. Left: heat sink of TO263; Right: solder pad layout of TO263.

with wire bonding technology. The whole component is covered with an epoxy resin except the terminals and the back side of the heat sink [7] which is also fixed by soldering to the PWB. The pad layout of the heat sink on the PWB is a metal surface perforated with thermal vias, whilst the contact surfaces of the heat sink are formed on the metal surface (marked with blocks in Fig. 2). In this way the heat loss is dissipated perpendicular through the PWB.

On these complicated solder pad layouts, the soldering process is very sensitive to the balance of the wetting force, therefore the occurrence of component displacement is more likely. It is very problematic to eliminate this type of failure during the manufacturing process. Therefore the goal of this project is to predict the component displacement by investigating the temperature distribution in the component. The chosen method is thermal simulations.

Generally, the temperature distribution is examined only at the level of assembled PWB. Usable 2D thermal models [1,2] and reflow thermal profile predictors [3] are already in use. These 2D models describe the temperature distribution of the assembled PWB, and can be used to predict and check the thermal profile of the reflow oven. However, for the description of temperature distribution at the level of discrete components, we need a 3D thermal model. In case of displacement a most important factor is the formation of joints under the heat sink. In this area we need a much finer resolution of the temperature distribution than is the case in other parts of the component (e.g. the epoxy body). Therefore we have developed a 3D thermal model using nonuniform cuboid grids that are concentrated to give the highest accuracy in this important region.

General purpose FEM analysis programs are able to employ a variety of element shapes and sizes in order to mesh the region of interest. However, our single purpose method described here is based on a nonuniform cuboid cell partition of elements in order to maintain simplicity and to enable high calculation speeds.

2. Model discretization

Focusing on the component level approach means that only the component and its direct environment is the only thing we need to consider in the assembly. Therefore the cuboid volume in Fig. 3 (marked by lines around the component) is the chosen objective area in the model and includes the partition of the PWB under the component.

The thermal (central) node theory [1,8] is applied to the model discretization which means that the objective area is divided into thermal cells. We have defined the basic thermal cell as a cuboid (Fig. 4). This shape is easy to handle and useful for our nonuniform grid. All cells contain a thermal node in their geometrical centre, which represents the thermal mass of the cell as a heat capacity [J/K]:

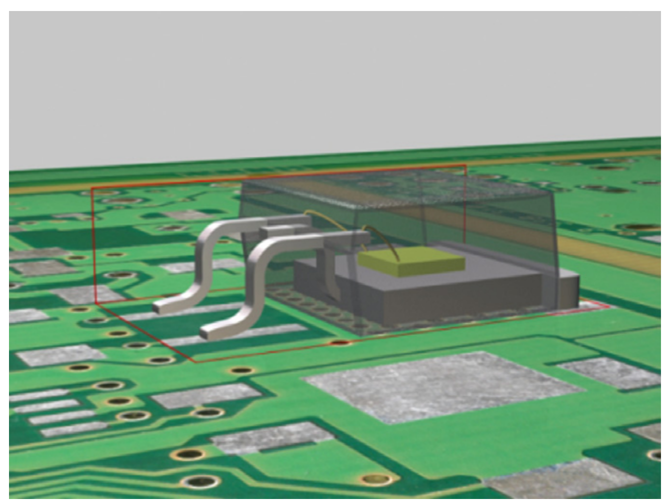

Fig. 3. The objective area of the model.

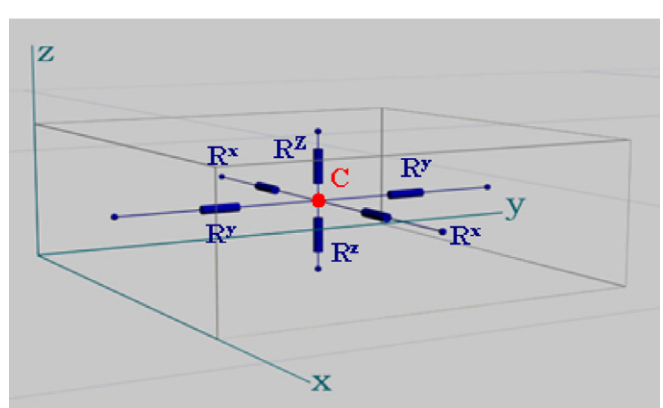

Fig. 4. The basic thermal cell.

$$C = C_S \cdot \rho \cdot V \quad (1)$$

where C_S is the specific heat capacity of the material [J/kg K], ρ is the density of the material [kg/m^3] and V is the volume of the cell [m^3]. The thermal conduction ability of the cells is described with thermal resistances [K/W] between the node and its cell borders:

$$R^x = \frac{l_x}{2 \cdot \lambda \cdot l_y \cdot l_z} \quad (2.1)$$

$$R^y = \frac{l_y}{2 \cdot \lambda \cdot l_x \cdot l_z} \quad (2.2)$$

$$R^z = \frac{l_z}{2 \cdot \lambda \cdot l_x \cdot l_y} \quad (2.3)$$

where l_x, l_y and l_z are the dimensions of the cell [m] and λ is the specific thermal conductivity of the material [W/m K].

According to the thermal node theory there should be thermodynamic equilibrium in all cells. Therefore during the cell partition, the borders of the different materials have to be taken into consideration. Our model contains the following types of cells: C1 epoxy resin (body of the component), C2 copper (heat sink and pins), C3 silicon (the chip), C4 N_2 gas (near the component). In addition some composites which are indivisible in our model range: C5 pure FR4 (epoxy resin + glass-fibre), C6 FR4 with metallization (metallization means the contact pads and the wiring as well) and C7 thermal vias. We have neglected the effect of the wire bonds because of their minimal mass.

Most of the physical properties of the cells are literature data from [9], but the thermal conductivity of C5, C6 and C7 are defined in different way because they highly depend on the construction of the applied FR4. The layer model [10,11] of our FR4 board (Fig. 5) is used for C5 and C6 which defines the lateral and vertical thermal conductivity in the following form:

$$\lambda_{\text{lat}} = \left(\sum_{i=1}^{M} \lambda_i \cdot t_i \right) / t \quad (3.1)$$

$$\lambda_{\text{ver}} = t / \left(\sum_{i=1}^{M} t_i / \lambda_i \right) \quad (3.2)$$

where t is the thickness of the FR4 board, λ_i is the thermal conductivity and t_i is the thickness of layer i. Our board contains 7 epoxy resin layers and 9 glass-fibre layers (and 1 copper layer in C6). In the case of epoxy resin and glass-fibre, the layer thicknesses are changing, therefore during the calculations we have used average values which are shown in Fig. 5. The λ parameters of the thermal

Table 1
Properties of the thermal cell types

Cell type	Material	C_s[J/kg k]	ρ[kg/m^3]	λ [W/m K]	
				Lateral (x, y)	Vertical (z)
C1	Epoxy resin	1100	1500	0.25	0.25
C2	Copper	385	8900	390	390
C3	Silicon	750	2300	120	120
C4	N_2[a]	0	0	0	0
C5	FR4	600	1900	0.76	0.53
C6	FR4 + metallization	570	2100	14.49	0.55
C7	Thermal via[b]	500	850	0.9	45

[a] This type is used only to detect of the solid–gas interface in the model. We do not consider them as real thermal cells.

[b] The equivalent density of the thermal via is only half of the FR4's density due to the trough hole.

vias mainly depend on the thickness of the hole metallization [12]. The applied values are determined according to the results of [12], based on the supposition that the thickness of the hole metallization is 10% compared to the radius of the hole. A summary of the physical properties of the defined thermal cell types can be seen in Table 1.

As we mentioned before we have used finer grid at the soldering surfaces than in the other parts of the objective area. The construction of the nonuniform grid is very important due to the implementation of the model. We used a simple but very useful condition for the faces of the thermal cells in the nonuniform grid: all neighbouring faces should have the same size. It ensures that all thermal cells can be in direct connection with maximum 6 other cells (all cells can have maximum 6 neighbours). In this case, all the neighbours are known, so the model can be implemented according to the indexing of the cells. Otherwise, (if the number of the neighbours is not maximized) the direct connections between the cells would be found by mathematical induction, so the implementation of the model would be more complicated.

This construction of the nonuniform grid can be achieved by the following steps: we define a basic uniform grid with section planes. The cell borders are the section lines of these planes. After this, adaptive interpolation (in the areas of interest) and decimation (in the less important areas) is done on the section planes separately along each axis. If the degrees of the interpolation and the decimation are the same then the number of cells will not change. A nonuniform grid can be seen in Fig. 6. In order to help a better

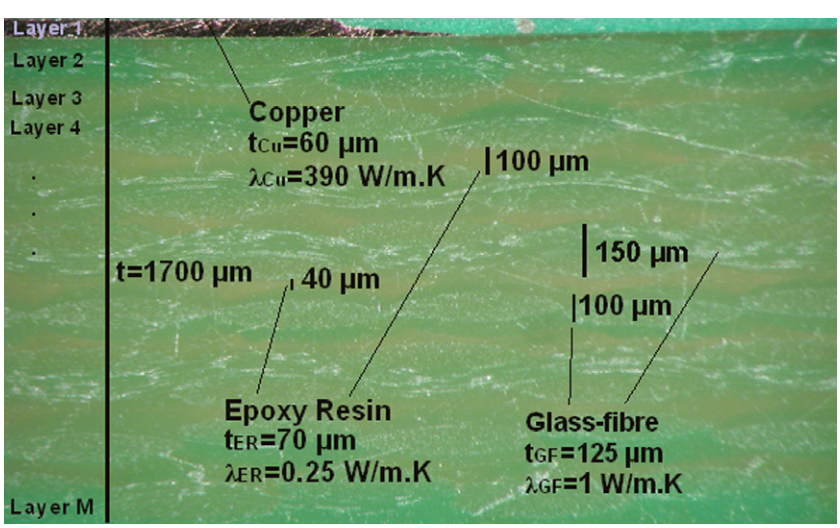

Fig. 5. Layer model of our FR4 board.

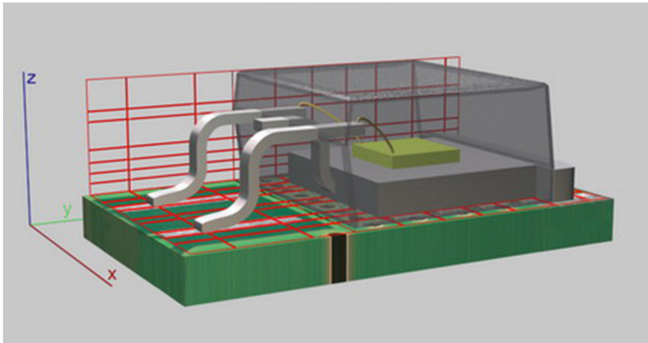

Fig. 6. Nonuniform resolution of the objective area.

visualization of this concept only some section lines of the section planes are visible in the picture. The nonuniform grid gives as more information from the important parts than the uniform with the same cell number.

The cell type and the physical position of the cells can be assigned by the coordinates of the intersections. The indexing of the cells is carried out by numbering the section planes (n_x, n_y and n_z according to the axis). All cells and their cell parameters can then be identified with these numbers: the heat capacity $C(n_x,n_y,n_z)$; the thermal resistances $R^x(n_x,n_y,n_z)$, $R^y(n_x,n_y,n_z)$ and $R^z(n_x,n_y,n_z)$ which also contain the direction of the resistance (Fig. 4).

3. Model description

The model is described by common heat conduction and convection equations and these are solved by the finite difference method in order to achieve high calculation speed and easy implementation. If the time steps are small enough then the boundary conditions and material properties can be assumed to remain constant over the time steps. The temperature change in the cells can be calculated by the backward Euler formula:

$$\frac{\mathrm{d}T_n(t)}{\mathrm{d}t} = \frac{T_n(t) - T_n(t-\mathrm{d}t)}{\mathrm{d}t} = \frac{F}{C} \tag{4}$$

where T_n is the temperature of the cell [K], t is the time [s], dt is the time step [s], F is the heat flow rate [W] and C is the heat capacity of the cell [J/K].

We use the following boundary conditions: there is only convection heat change between the solid–gas boundaries, and the convection heat always flows from the gas to the solid (the gas is always hotter than the solid). The solid–solid boundaries of the model (these are located only in the FR4 board) are considered to be adiabatic. The heat spreads by conduction heat transport between the thermal cells in the model. In forced convection reflow ovens, we neglect the radiation heat because it is very small compared to the convection or conduction heat. Nevertheless the latent heat of the solder is also negligible [1–3], it is only ~60 J/g.

The convective heat flow rate is

$$F_C(t,\underline{r}) = S \cdot \alpha(\underline{r}) \cdot (T_\mathrm{h} - T_n(t)) \tag{5}$$

where $\alpha(\underline{r})$ is the heat transfer coefficient [W/m^{2}K] depending on the location, S is the heated surface [m^2] and T_h is the heater gas temperature [K] (which is considered to be constant in the ambience of the modelled component in a given heater zone). Conduction heat flow is caused by the temperature difference between the cells:

$$F_K(t) = \frac{T_{n-1}(t) - T_n(t)}{R_{n-1} + R_n} \tag{6}$$

where R_{n-1} and R_n are the thermal resistances between the adjacent cells, whilst $T_{n-1}(t)$ and $T_n(t)$ are the temperatures of the adjacent cells.

We assume the following initial condition $T_1(0) \cong T_2(0) \cong \ldots \cong T_n(0)$ meaning that, before the heating, the temperature gradient of our system is very small. This assumption is always valid for equilibrium thermodynamics systems. Therefore, between the adjacent cells $F_K(0) \cong 0$, this means that the conduction heat is generated only by convection heating. Using (4)–(6), this is the general form of the differential equation which describes the temperature change of the cells as the following:

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{C} \cdot \left[\sum_{i=0}^{k} F_{C_{(i)}} + \sum_{j=1}^{6-k} F_{K(j)}\right] \tag{7}$$

where k is the number of the solid–gas interfaces and $6-k$ is the number of adjacent cells.

For the description of the whole model, a differential equation system is needed. With an appropriate selection of the cells parameters and initial assumptions, the differential equation system can be easily generated. The indexing method, shown in Section 2, will be our starting point. All cells have an own heat capacity, six faces and six thermal resistances from the six faces. The heat capacities can be handled as values $C(n_x,n_y,n_z)$ but the surfaces and thermal resistances are to be stored in vectors. Although all cells have only three different surfaces, these are stored in six element vectors so it will be useful for general implementation:

$$\underline{S}(n_x,n_y,n_z) = \begin{bmatrix} S^x(n_x,n_y,n_z) \\ S^{-x}(n_x,n_y,n_z) \\ S^y(n_x,n_y,n_z) \\ S^{-y}(n_x,n_y,n_z) \\ S^z(n_x,n_y,n_z) \\ S^{-z}(n_x,n_y,n_z) \end{bmatrix} \tag{8}$$

where $S^x(n_x,n_y,n_z) = S^{-x}(n_x,n_y,n_z)$, these are opposite faces of the cell which are perpendicular to x axis, etc. For the general implementation the thermal resistances should be transformed to thermal conductances. The $\underline{G}(n_x,n_y,n_z)$ vectors have the following form:

$$\underline{G}(n_x,n_y,n_z) = \begin{bmatrix} G^x(n_x,n_y,n_z) \cdot G^x(n_x-1,n_y,n_z)/(G^x(n_x,n_y,n_z) + G^x(n_x-1,n_y,n_z)) \\ G^x(n_x,n_y,n_z) \cdot G^x(n_x+1,n_y,n_z)/(G^x(n_x,n_y,n_z) + G^x(n_x+1,n_y,n_z)) \\ G^y(n_x,n_y,n_z) \cdot G^y(n_x,n_y-1,n_z)/(G^y(n_x,n_y,n_z) + G^y(n_x,n_y-1,n_z)) \\ G^y(n_x,n_y,n_z) \cdot G^y(n_x,n_y+1,n_z)/(G^y(n_x,n_y,n_z) + G^y(n_x,n_y+1,n_z)) \\ G^z(n_x,n_y,n_z) \cdot G^z(n_x,n_y,n_z-1)/(G^z(n_x,n_y,n_z) + G^z(n_x,n_y,n_z-1)) \\ G^z(n_x,n_y,n_z) \cdot G^z(n_x,n_y,n_z+1)/(G^z(n_x,n_y,n_z) + G^z(n_x,n_y,n_z+1)) \end{bmatrix} \tag{9}$$

According to the construction of the model, multiplying by zero is carried out in the positions of solid-gas interfaces. These elements of $\underline{G}(n_x,n_y,n_z)$ are zero.

As we have stated earlier, the convection efficiency could be changed on the different faces of the components [4–6]. We have applied the flowing assistant vectors:

$$\underline{\alpha}(n_x,n_y,n_z) = \begin{bmatrix} \alpha^x(n_x,n_y,n_z) \\ \alpha^{-x}(n_x,n_y,n_z) \\ \alpha^y(n_x,n_y,n_z) \\ \alpha^{-y}(n_x,n_y,n_z) \\ \alpha^z(n_x,n_y,n_z) \\ \alpha^{-z}(n_x,n_y,n_z) \end{bmatrix} \quad (10)$$

where $\alpha^x(n_x,n_y,n_z)$ is the heat transfer coefficient on $S^x(n_x,n_y,n_z)$ and $\alpha^{-x}(n_x,n_y,n_z)$ is the heat transfer coefficient on $S^{-x}(n_x,n_y,n_z)$, etc. The elements of $\underline{\alpha}(n_x,n_y,n_z)$ have to be zero at those positions where $G(n_x,n_y,n_z)$ are not zero. This means that $\underline{\alpha}(n_x,n_y,n_z)$ are zero on those positions where adjacent cells are located near the given cell. To achieve a compact description, the temperature differences between a given cell and its adjacent cells are also collected into assistant vectors:

$$\underline{T_K}(n_x,n_y,n_z) = \begin{bmatrix} T(n_x-1,n_y,n_z) - T(n_x,n_y,n_z) \\ T(n_x+1,n_y,n_z) - T(n_x,n_y,n_z) \\ T(n_x,n_y-1,n_z) - T(n_x,n_y,n_z) \\ T(n_x,n_y+1,n_z) - T(n_x,n_y,n_z) \\ T(n_x,n_y,n_z-1) - T(n_x,n_y,n_z) \\ T(n_x,n_y,n_z+1) - T(n_x,n_y,n_z) \end{bmatrix} \quad (11)$$

We need only those values of $\underline{T_K}(n_x,n_y,n_z)$ where adjacent cells exist. Therefore $\underline{T_K}(n_x,n_y,n_z)$ vectors will be filtered by the $\underline{G}(n_x,n_y,n_z)$ vectors. In those positions where the $\underline{G}(n_x,n_y,n_z)$ vector is zero, the value of the $\underline{T_K}(n_x,n_y,n_z)$ will be disregarded. The solid–gas interfaces are chosen by $\alpha(n_x,n_y,n_z)$. By the application of the assistant vectors (8)–(11) this gives us the general expression of the differential equation system which best describes the model:

$$\frac{\mathrm{d}T(n_x,n_y,n_z)}{\mathrm{d}t} = \begin{bmatrix} \frac{\underline{\alpha}(n_x,n_y,n_z)^T \cdot \underline{S}(n_x,n_y,n_z) \cdot (T_h - T(n_x,n_y,n_z))}{C(n_x,n_y,n_z)} + \\ + \frac{\underline{G}(n_x,n_y,n_z)^T \cdot \underline{T_K}(n_x,n_y,n_z)}{C(n_x,n_y,n_z)} \end{bmatrix} \quad (12)$$

If we do not maximize the number of the neighbours then the assistant vectors (8)–(11) would be matrices with $N \times N$ dimensions (N is the cell number), therefore Eq. (12) and its implementation would be more complex.

4. Results and discussion

The model was implemented using MATLAB 7.0 software. Verification of our model was achieved by temperature measurement and simulations using Comsol 3.3 which is widely used general purpose FEM analyzer. We have defined a nonuniform grid with 792 thermal cells (the applied resolution the same as in Fig. 6) and K-type thermocouple was used for the measurements. The investigation itself took place in the 6th heater zone of the oven (where the melting of the solder occurs) with a thermal profile for leaded solders. (Although the study was done using leaded solders, our model is absolutely suitable for lead-free solders). The initial condition was $T(0) = 178.5$ °C for the temperature of the objective area. The heater temperature was $T_h = 225$ °C, and the time of heating was 20 s.

In forced convection reflow ovens, the heat transfer coefficient highly depends on the location and position of the heated surface. This is caused by the gas flow circumstances in the oven and the construction of the oven. In our earlier works we have examined the 3D direction characteristics of the heat transfer coefficient in reflow ovens [5,6]. Our oven gave us the following values (Table 2) which we applied during the simulations.

Two thermal cells were chosen from the model, their temperature changes were measured and calculated using our model and the Comsol 3.3. The examined cell types were: C1 (from the right lower corner of the epoxy body) and C2 (from the left upper corner of the heat sink). The measured values and calculated data are compared in Fig. 7. The verification results prove that the convergence of our model with that of the measured data is very good. Although our approach is completely different, it can also be seen that the calculated results are almost identical.

Comparing the abilities of our model with that of the Comsol 3.3 gave us the following results. The data entry and the generation of the model took nearly the same time in both systems, but the calculation in our model was much faster than in the general purpose FEM analyzer. Tested on the same hardware configuration,

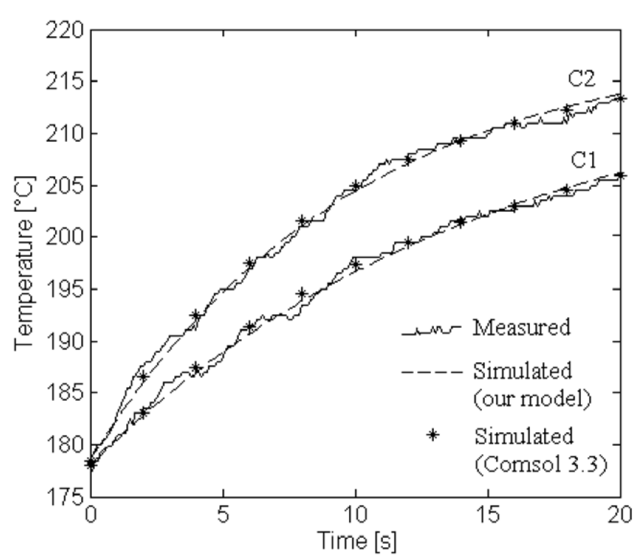

Fig. 7. Verification of the model.

Table 2
Heat transfer coefficients

At h height from the assembly	α [W/m²K] (on the different interfaces)					
	α^x	α^{-x}	α^y	α^{-y}	α^z	α^{-z} [a]
1 mm	51.21	32.89	94.40	100.35	41.61	38.64
3 mm	88.81	55.35	127.45	138.88	55.06	50.26
6 mm	84.04	47.75	105.08	115.66	67.85	62.95
9 mm	74.92	40.15	100.57	108.56	79.44	74.29
12 mm	68.74	36.09	97.60	104.87	89.88	82.74
15 mm	64.21	33.17	93.53	99.19	92.16	88.72

[a] In this case the h means the same but with negative values.

the calculation time was less than 3 s using our model, while the Comsol 3.3. took more than 52 s. In addition, our model proved to be much easier to use than the general purpose FEM analyzers.

We also examined the stability of our model. Using the objective area in Fig. 3, the model was stable with up to 5000 cells, each cell size approximately $\sim$0.18 mm^3. However, as we will see, in our application $\sim$1 mm^3 cell size ($\sim$800 cells) is more than enough. Besides, the stability of the model is highly dependent on the satisfactory performance of the mathematical software (in this case MATLAB 7.0). We observed during the tests that the calculation time increased linearly with the cell numbers (as expected) in line with the model description (Section 3) and Eq. (12).

The displacement of TO263 package (Fig. 1.) is investigated by means of a thermal simulation. The displacement can be caused by the heat transfer coefficient differences around the component. In the mechanism, the heating deviation results in a time difference between the starting of the melting process on different parts of the soldering surfaces. This breaks the balance of the wetting force and increases the chances that the component will skew. As mentioned in Section 1, when displacement occurs the most important thing is the joint formation under the heat sink, the largest temperature differences can occur between contact surfaces. Therefore we have studied in detail the temperature distribution of this area.

The applied nonuniform grid is the same as in the case of verification (Fig. 6), the $x-y$ projection in the contact surfaces can be seen in (Fig. 8). In this grid, the 13 contact surfaces are described by 31 thermal cells (Fig. 8). But the cells which represent the same contact surface can be dealt with as one. The examined cell groups are shown as squares in Fig. 8. We have applied the heat transfer coefficient from Table 2, and investigated the case when the y direction in our model (Fig. 8) is parallel with the lateral walls of the oven. In this position of the component, there is considerable heat transfer coefficient deviation between right and left faces of the component (Fig. 8). Therefore, in this case the displacement will occur most likely along the x axis.

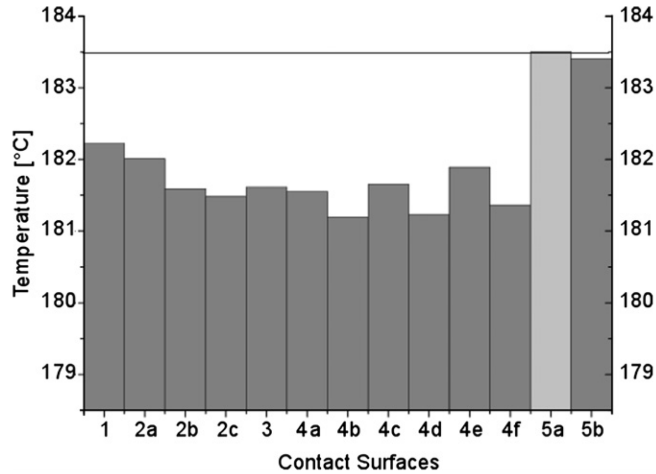

Fig. 9. Temperature distribution of the contact surfaces, $t = 1.90$ s.

The simulation started when the assembly was entered in the 6th heater zone, with the initial condition $T_1(0) \cong T_2(0) \cong \ldots \cong T_n(0) = 178°C$. This is the starting temperature of the 6th heater zone. We observed that the temperature gradient of the component becomes very small by the time it reaches the exit of the heater zones and it remains also small until the entrance of the following heater zone. This also can be seen in the verification results at $t = 0$ s (Fig. 7). Therefore our initial condition is close to reality.

In the case of eutectic leaded solders, the melting and the wetting starts close to 183.5 °C; therefore during the simulation we did a detailed search to see when it was happening on the different contact surfaces. The applied time step was dt = 10 ms. The first melting occurred on the contact surface 5a, at $t = 1.90$ s (Fig. 9). At this time the temperature distribution on the contact surfaces of the heat sink was nearly homogeneous. Less than 1 s later (Fig. 10), higher temperature differences occurred between the contact surfaces. From the six contact surfaces on which the solder had melted, five were located on the right side of the solder pad layout (Fig. 8). The temperature distribution should be studied along the symmetry axis of the component (which is the normal alignment), to the expected direction of the displacement between

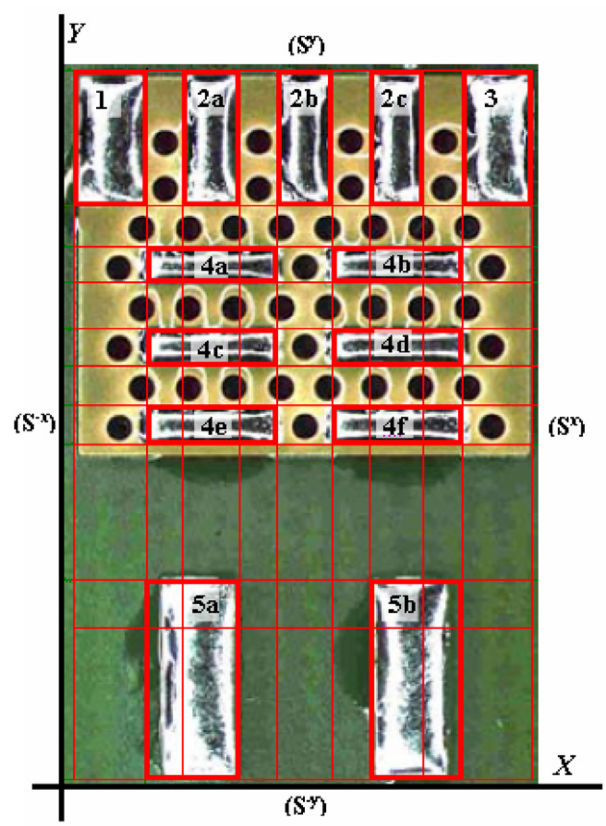

Fig. 8. $x-y$ projection of the applied nonuniform grid.

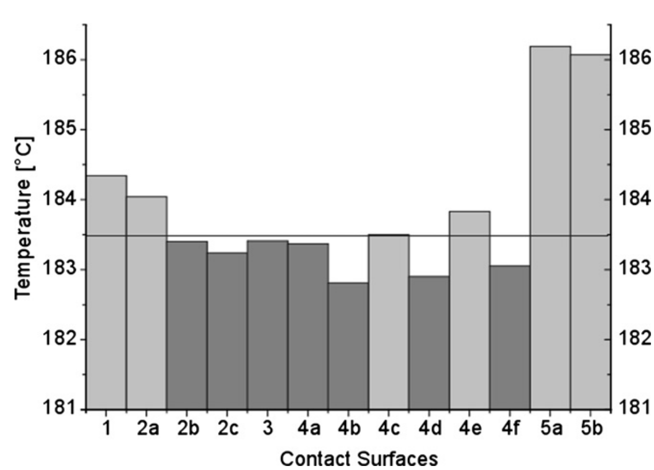

Fig. 10. Temperature distribution of the contact surfaces, $t = 2.81$ s.

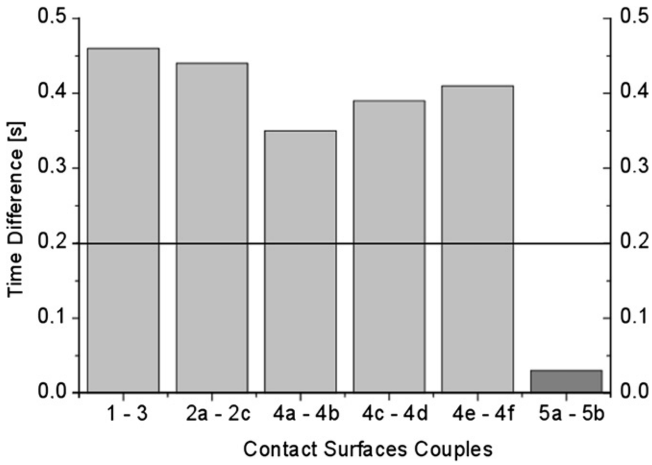

Fig. 11. Melting time differences between the opposite contact surfaces.

the adequate contact surface couples. According to the industrial results, if the time difference between the starting of the melting on the different contact surfaces is larger than 0.2 s the displacement of the component can occur [13,14]. As our results show (Fig. 11), the heat transfer coefficient deviation between the right and left faces of the component will cause melting and wetting under the heat sink to happen nearly 0.4 s faster on the right side in (Fig. 8). This can result in the displacement of the component in the *x* direction.

5. Conclusions

We have shown that a relatively simple method as the thermal node theory can be a useful tool for investigating a complex problem such as component displacement. Using adaptive interpolation and decimation our model can improve the accuracy of the interested areas without increasing their complexity. The verification results prove that the convergence of our model with that of measured data is very favourable. Although the solution is completely different, the calculated results from a general FEM analyzer and from our model are almost identical. However, the time taken for calculation by our model is very short (only some seconds) when compared with the general FEM analyzers. The development of the nonuniform grid according to the interested and uninterested parts of the objective area takes the most time to construct in the simulation. How much depends on the experience of the user!

Although we have only showed results from one investigation of a TO263 package displacement, our model is able to predict most kinds of displacement and movement of SMDs during reflow soldering if they are caused by inhomogeneous thermal properties of the components and/or inhomogeneous convection efficiency. Further work is required to test the limits of the accuracy in the function of cell partition resolution and how to handle effectively the conditions at the border of the objective area. In addition, the modelling approach suggested in this paper, is also applicable for simulation and optimization in other thermal processes. For example, where the inhomogeneous convection heating or conduction properties can cause problems, such as the preheating system of wave and selective soldering machines, high temperature surface cleaning and dye-drying process.

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